Type MS · Angles and areas of triangles inside a solid

Answer 1 of 8 · Question 6 part b · 2 marks

2023 NOV · Paper 01 · extract 1 of 4 · find part b in this mark scheme extract

6(a)	$(AC=)\sqrt{x^2+x^2}\ (=\sqrt{2}x)$ **or** $(AC=)\dfrac{x}{\sin 45}$ **or** $(AC=)\dfrac{x}{\cos 45}$ $(AO=)\sqrt{\left(\dfrac{x}{2}\right)^2+\left(\dfrac{x}{2}\right)^2}\ \left(=\sqrt{\dfrac{x^2}{2}}=\dfrac{x}{\sqrt{2}}\right)$ **or** $(AO=)\dfrac{\frac{x}{2}}{\sin 45}$ **or** $(AO=)\dfrac{\frac{x}{2}}{\cos 45}$ $(VO=)\sqrt{x^2-\left(\dfrac{"\sqrt{2}x"}{2}\right)^2}$ **or** $\sqrt{x^2-\left(\dfrac{1}{2}\sqrt{x^2+x^2}\right)^2}$ **or** $\sqrt{x^2-\dfrac{x^2}{2}}\ \left(=\sqrt{\dfrac{1}{2}x^2}\right)$ **or** $(VO=)\dfrac{\sqrt{2}}{2}\tan 45$ $=\dfrac{\sqrt{2}}{2}x$ (cm)*	M1 M1 A1*cso [3]
ALT	Any indication the candidate realises triangle AVC is right angled and isosceles. $\sin 45=\dfrac{VO}{x}$ **or** $x\sin 45=VO$ $VO=\dfrac{\sqrt{2}}{2}x$ *	M1 M1 A1*cso [3]
(b)	$(\angle AVC=)\cos^{-1}\left(\dfrac{x^2+x^2-\left(\sqrt{2}x\right)^2}{2\times x\times x}\right)$ **or** $(\angle AVC=)2\times\sin^{-1}\left(\dfrac{\frac{\sqrt{2}}{2}x}{x}\right)$ **or** $(\angle AVC=)2\times\cos^{-1}\left(\dfrac{\frac{\sqrt{2}}{2}x}{x}\right)$ **or** $(\angle AVC=)2\times\tan^{-1}\left(\dfrac{\frac{\sqrt{2}}{2}x}{\frac{\sqrt{2}}{2}x}\right)$ $=90^\circ$	M1 A1 [2]

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(c)	Let M be the midpoint of AB and let N be the midpoint of DC	
	$(VM = VN =)\sqrt{x^2 - \left(\frac{x}{2}\right)^2} = \left(\frac{\sqrt{3}}{2}x\right)$	M1
	$\cos MVN = \dfrac{\left("\frac{\sqrt{3}}{2}x"\right)^2 + \left("\frac{\sqrt{3}}{2}x"\right)^2 - x^2}{2\times\left("\frac{\sqrt{3}}{2}x"\right)\times\left("\frac{\sqrt{3}}{2}x"\right)}$ **or**	
	$(\angle MVN =)\cos^{-1}\left(\dfrac{\left("\frac{\sqrt{3}}{2}x"\right)^2 + \left("\frac{\sqrt{3}}{2}x"\right)^2 - x^2}{2\times\left("\frac{\sqrt{3}}{2}x"\right)\times\left("\frac{\sqrt{3}}{2}x"\right)}\right)$ **or**	M1
	$(\angle MVN =)2\times\cos^{-1}\left(\dfrac{\frac{\sqrt{2}}{2}x}{"\frac{\sqrt{3}}{2}x"}\right)$ **or** $(\angle MVN =)2\times\sin^{-1}\left(\dfrac{\frac{1}{2}x}{"\frac{\sqrt{3}}{2}x"}\right)$	
	or $(\angle MVN =)2\times(\angle MVO =)$ **and** $\cos MVO = \dfrac{\frac{\sqrt{2}}{2}x}{"\frac{\sqrt{3}}{2}x"}$	
	$70.528...^\circ \approx 70.5^\circ$	A1 [3]
	ALT	M1
	$OM = \frac{x}{2}$	
	$(\angle MVN =)2\tan^{-1}\left[\dfrac{\frac{x}{2}}{\frac{\sqrt{2}x}{2}}\right] = \left(2\tan^{-1}\left[\frac{1}{\sqrt{2}}\right] =\right)70.528...^\circ \approx 70.5^\circ$	M1A1 [3]
(d)	Volume $= \frac{1}{3}\times\left(\frac{\sqrt{2}}{2}x\right)\times x^2\left(=\frac{\sqrt{2}}{6}x^3\right)$ oe	M1
	$200 = \frac{1}{3}\times\left(\frac{\sqrt{2}}{2}x\right)\times x^2 \Rightarrow x =$	dM1
	$x = 9.4672... \approx 9.47(\text{cm})$	A1 [3]
		Total 11 marks

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Part	Mark	Notes
For part (a), we must see the algebraic proof, including x throughout.		
(a)	M1	For correct use of Pythagoras theorem or any correct appropriate trigonometry and a complete method to find the length AC or AO ie candidates must rearrange correctly – minimum steps as shown in mark scheme and no errors must be present.
	M1	For a full, correct and complete method to find VO, using their AC/AO.
	A1 cso	For $\frac{\sqrt{2}}{2}x$ (cm) – correct solution only, no errors or omissions, minimum steps as shown.
ALT	M1	Any indication the candidate realises triangle AVC is right angled and isosceles. This mark may be implied by correct working for the next method mark or present on the diagram.
	M1	For the correct trig minimum steps as shown and no errors.
	A1 cso	For $\frac{\sqrt{2}}{2}x$ (cm) – correct solution only, no errors or omissions.
(b)	M1	For any correct appropriate trigonometry to find angle AVC. $(\angle AVC =)$ If finding $\angle AVO$, must be doubled for this mark. AC must be $\sqrt{2}x$ oe, AO must be $\frac{\sqrt{2}}{2}x$ oe.
	A1	For 90°
	Allow M1A1 for candidates writing down the correct answer by inference.	
Consistent omission of x is permissible for all marks in part (b).		
(c)	M1	For correct Pythagoras theorem to find the lengths of VM or VN
	M1	For any correct trigonometry to find angle MVN, such as the statements shown in the mark scheme. The candidate can state, for example, cos(MVN) = fully correct expression (using their VN and VM) or, for example, $\cos^{-1}$(fully correct expression), where they do not have to write angle MVN = i.e – this can involve any fully correct method to reach an equation which would allow them to use inverse trig to find MVN or if they use inverse trig, a fully correct expression must be seen with the correct inverse trig function. Send any items where allocation of marks is unclear (potentially using other methods to the main and ALT) to Review. or fully correct **method** to find angle MVO which must then be doubled for this mark.
	A1	For any answer stated which rounds to 70.5° (ie more decimal places is condoned).
Consistent omission of x is permissible for all marks in part (c).		
ALT	M1	States or uses $OM = \frac{x}{2}$
	M1	For fully correct **method** to find angle MVO which must then be doubled for this mark.
	A1	For any answer stated which rounds to 70.5° (ie more decimal places is condoned).
(d)	M1	For a correct unsimplified expression for the volume of the pyramid as shown.
	dM1	For placing their expression = 200, which must involve a term in x^3, and a fully correct method/rearrangement to solve. Note this mark is dependent on the first method mark, so they must have got the correct unsimplified expression even if then simplified incorrectly.
	A1	For any final answer stated which rounds to 9.47 (ie more decimal places is condoned).

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Answer 1 of 8 · Question 6 part b · 2 marks

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Question	Scheme	Marks

Type MS · Angles and areas of triangles inside a solid

Answer 2 of 8 · Question 8 part d · 6 marks

2019 JUN · Paper 02 (regional R) · extract 1 of 3 · find part d in this mark scheme extract

8 (a)	$BN = 10\cos 60^\circ$ $= 5$ (cm)*	M1 A1cso [2]
(b)	$FN^2 = 10^2 - 5^2 = 75$ or $FN = 10\sin 60 = 5\sqrt{3}$ or $FN = 5\tan 60$ $NE = \sqrt{12^2 + 75} = 14.7986... \approx 14.8$ (cm)	M1 M1A1 [3]
(c)	Let R be the midpoint of FN and P be the point on AD that corresponds to point N on BC. $\angle RPN = \tan^{-1}\left(\dfrac{\sqrt{75}/2}{12}\right) = 19.84167...^\circ \approx 19.8^\circ$	M1A1 [2]
(d)	Let the perpendicular from Y to line BC meet BC at Z $AY = \sqrt{AZ^2 + ZY^2} = \sqrt{12^2 + 6.5^2 + \left(\dfrac{\sqrt{75}}{2}\right)^2} = \sqrt{205}$ $FC = \sqrt{FN^2 + NC^2} = \sqrt{75 + 3^2} = \sqrt{84} \Rightarrow FY = \sqrt{21}$ So $EY = \sqrt{12^2 + 21} = \sqrt{165}$ $\angle AYE = \cos^{-1}\left(\dfrac{205 + 165 - 100}{2 \times \sqrt{205} \times \sqrt{165}}\right) = 42.7745...^\circ \approx 42.8^\circ$	M1A1 M1 M1 M1A1 [6]
		Total 13 marks

(a)	M1	Use cosine or sine in triangle BNF, or any complete method. $BN = 10\cos 60^\circ$ or $\cos 60^\circ = \dfrac{BN}{10}$ or $\dfrac{BN}{\sin 30} = \dfrac{10}{\sin 90}$ or $BN = 10\sin 30$ or $FN = 10\sin 60 = 5\sqrt{3}$ and $BN^2 = 10^2 - (5\sqrt{3})^2$ (may use any single letter instead of BN)
	A1 cso	$BN = 5$ (cm) Correct answer from correct working. (Answer given.) Must show a calculation in the form $BN = \ldots$ e.g. $BN = 10\cos 60$
(b)	M1	Use Pythag or trig to find a numerical expression for FN or FN^2 eg $FN^2 = 10^2 - 5^2$ $(= 75)$ or $FN = 10\sin 60$ $(= 5\sqrt{3} = 8.66...)$ or $FN = 5\tan 60$
	M1	Use Pythag with their FN to find a numerical expression for NE, eg $NE = \sqrt{12^2 + (5\sqrt{3})^2}$ $(= \sqrt{219})$
	A1	$NE = 14.8$ (cm) **given to 3SF**.
(c)	M1	Use tangent with their FN, eg $\tan RPN = \left(\dfrac{\frac{1}{2}\sqrt{75}}{12}\right)$ or any complete method.
	A1	19.8° **given to 1DP**.

Type MS · Angles and areas of triangles inside a solid

Answer 2 of 8 · Question 8 part d · 6 marks

2019 JUN · Paper 02 (regional R) · extract 2 of 3 · find part d in this mark scheme extract

(d)	M1	Complete method to find AY eg $AY = \sqrt{(12^2 + 6.5^2) + \left(\frac{\sqrt{75}}{2}\right)^2}$ $(= \sqrt{205} = 14.31...)$ (allow their FN)
	A1	$AY = \sqrt{205}$ $(= 14.31\ldots)$ (may be implied by subsequent working)
	M1	Complete method to find FY eg $FC = \sqrt{75 + 3^2}$ $\left(= \sqrt{84}\right) \Rightarrow FY = \frac{1}{2}\sqrt{84}$ $(= \sqrt{21})$ (allow their FN) eg $FC = \sqrt{8^2 + 10^2 - 2 \times 8 \times 10 \cos 60}$ $\left(= \sqrt{84}\right) \Rightarrow FY = \frac{1}{2}\sqrt{84}$ $(= \sqrt{21} = 4.58...)$
	M1	Method to find EY using their FY (correct or from a correct method) eg $EY = \sqrt{12^2 + 21}$ $(= \sqrt{165} = 12.84...)$
	M1	Substitute their values for AY and EY into the cosine rule used in triangle AYE. $\cos AYE = \left(\frac{205 + 165 - 100}{2 \times \sqrt{205} \times \sqrt{165}}\right)$ oe
	A1	$\angle AYE = 42.8^\circ$ **given to 1DP**. Allow more than 1DP if 1DP was penalised in part (c).

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Answer 2 of 8 · Question 8 part d · 6 marks

2019 JUN · Paper 02 (regional R) · extract 3 of 3 · find part d in this mark scheme extract

Question number	Scheme	Marks

Type MS · Angles and areas of triangles inside a solid

Answer 3 of 8 · Question 4 part b · 2 marks

2022 JAN · Paper 02 · extract 1 of 3 · find part b in this mark scheme extract

4(a)	$FC = \sqrt{10^2 + 10^2 + 10^2} = \sqrt{300}$	M1A1 [2]
(b)	$\cos FCA = \dfrac{\sqrt{200}}{\sqrt{300}} \Rightarrow \angle FCA = 35.3^\circ$	M1A1 [2]
(c)	$CX = \dfrac{\sqrt{200}}{2} = \sqrt{50}$, $FX = \sqrt{50+100} = \sqrt{150}$ $\cos \angle FXC = \dfrac{'150'+'50'-'300'}{2\times\sqrt{'150'}\times\sqrt{'50'}} \Rightarrow \angle FXC = 125.2643...^\circ \Rightarrow$ awrt 125°	M1,M1A1 M1A1 [5]
	Total	**9 marks**

Part	Mark	Notes
(a)	M1	For using Pythagoras theorem or any appropriate trigonometry to find FC. $FC = \sqrt{10^2 + 10^2 + 10^2} = ...$ OR $AC = \sqrt{10^2 + 10^2} = \sqrt{200}$ $FC = \sqrt{10^2 + 200} = \sqrt{300}$
	A1	For the correct exact length of FC = $\sqrt{300}\left(=10\sqrt{3}\right)$
(b)	M1	For using any appropriate trigonometry to find the required angle. They must find a value for the award of this mark. $\cos \angle FCA = \dfrac{\sqrt{200}}{\sqrt{300}} \Rightarrow \angle FCA = \left(35.2643^\circ\right)$, $\sin \angle FCA = \dfrac{10}{\sqrt{300}} \Rightarrow \angle FCA = \left(35.2643^\circ\right)$, $\tan \angle FCA = \dfrac{10}{\sqrt{200}} \Rightarrow \angle FCA = \left(35.2643^\circ\right)$ Allow awrt 14.1 for $\sqrt{200}$ and awrt 17.3 for $\sqrt{300}$
	A1	$\angle FCA = 35.2643...^\circ \approx 35.3^\circ$ (awrt) Accept answers which round to 35.3°, but they must round to this value.

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Answer 3 of 8 · Question 4 part b · 2 marks

2022 JAN · Paper 02 · extract 2 of 3 · find part b in this mark scheme extract

(c)	M1	Let X be the midpoint of BH. For any method (Pythagoras or trigonometry) to find the length CX $CX = \frac{\sqrt{200}}{2} = \left(\sqrt{50} \text{ or } 5\sqrt{2}\right)$
	M1	For any method (Pythagoras or trigonometry) to find the length FX $FX = \sqrt{50+100} = \left(\sqrt{150} \text{ or } 5\sqrt{6}\right)$ or $FX = \sqrt{200-50} = \left(\sqrt{150} \text{ or } 5\sqrt{6}\right)$
	A1	For both correct lengths $\sqrt{50}$ and $\sqrt{150}$
	M1	For the correct cosine rule to find angle FXC using their lengths $\cos\angle FXC = \frac{'150'+'50'-'300'}{2\times\sqrt{'150'}\times\sqrt{'50'}} \Rightarrow \angle FXC = \left(125.2643...^{\circ}\right)$ NB: Check that the cosine rule they are using is to find the required angle.
	A1	For the correct angle with awrt. $\angle FXC = 125^{\circ}$ Accept answers which round to 125°, but they must round to this value.
	ALT – Uses the right-angled triangle *FGX*	
	M1	Let X be the midpoint of BH. For any method (Pythagoras or trigonometry) to find the length GX $GX = \frac{\sqrt{200}}{2} = \left(\sqrt{50} \text{ or } 5\sqrt{2} \text{ or awrt } 7.07\right)$ If they use the sin ratio to find the angle, identify FG = 10 (cm)
	M1	For any method (Pythagoras or trigonometry) to find the length FX $FX = \sqrt{50+100} = \left(\sqrt{150} \text{ or } 5\sqrt{6} \text{ awrt } 12.2\right)$ or $FX = \sqrt{200-50} = \left(\sqrt{150} \text{ or } 5\sqrt{6} \text{ awrt } 12.2\right)$ **OR** If they use the tan ratio to find the angle, identify FG = 10 (cm)
	A1	For both correct lengths $\sqrt{50}$ and 10 for tan **or** $\sqrt{50}$ and $\sqrt{150}$ for cos **or** $\sqrt{150}$ and 10 for sin Allow decimal equivalents, i.e., 7.07 and 12.2
	M1	$\cos\angle FXG = \frac{\sqrt{50}}{\sqrt{150}} \Rightarrow \angle FXG = \left(54.7356...^{\circ}\right)$, $\sin\angle FXG = \frac{10}{\sqrt{150}} \Rightarrow \angle FXG = \left(54.7356...^{\circ}\right)$, $\tan\angle FXG = \frac{10}{\sqrt{50}} \Rightarrow \angle FXG = \left(54.7356...^{\circ}\right)$ So the required angle is $\angle FXC = 180 - '54.7356...'^{\circ} = \left(125.264...^{\circ}\right)$ We must see this calculation for the award of this mark. Do **not** award just for 54.7356…°
	A1	For the correct angle with awrt. $\angle FXC = 125^{\circ}$ Accept answers which round to 125°, but they must round to this value.

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Answer 3 of 8 · Question 4 part b · 2 marks

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Question	Scheme	Mark

Type MS · Angles and areas of triangles inside a solid

Answer 4 of 8 · Question 10 part c · 2 marks

2024 JUN · Paper 02 (regional R) · extract 1 of 3 · find part c in this mark scheme extract

10(a)	$AC = \sqrt{(2x)^2 + x^2} = (\sqrt{5}x)$ $VA = '\sqrt{5}x' \sin 45^o = \frac{\sqrt{10}}{2}x$ *	B1 M1A1 cso [3]
(b)	Height $= \sqrt{\left(\frac{\sqrt{10}}{2}x\right)^2 - \left(\frac{\sqrt{5}}{2}x\right)^2} = \frac{\sqrt{5}}{2}x$	M1A1 [2]
(c)	$\cos \angle VBA = \frac{1}{\frac{\sqrt{10}}{2}}$ $\Rightarrow \angle VBA = 50.768...^o \approx 50.8^o$ awrt	M1 A1 [2]
(d)	Let M be the point at which the diagonals AC and BD meet. The required angle is either AMB or DMC AM = MB $= \frac{AC}{2} = \frac{\sqrt{5}x}{2}$ or DM=MC=$\frac{AC}{2} = \frac{\sqrt{5}x}{2}$ $\cos \angle AMB = \frac{\left(\frac{\sqrt{5}}{2}\right)^2 + \left(\frac{\sqrt{5}}{2}\right)^2 - 2^2}{2 \times \frac{\sqrt{5}}{2} \times \frac{\sqrt{5}}{2}}$ oe $(\cos \angle AMB = -\frac{3}{5} \Rightarrow \angle AMB = 126.869...^o) \approx 126.9^o$ awrt	B1 B1 M1 A1 [4]
(e)	$9\sqrt{5} = \frac{1}{3} \times 2x \times x \times "\frac{\sqrt{5}}{2}x" \Rightarrow x = \cdots$ $x = 3$	M1 A1 [2]
		Total 13 marks

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Answer 4 of 8 · Question 10 part c · 2 marks

2024 JUN · Paper 02 (regional R) · extract 2 of 3 · find part c in this mark scheme extract

Part	Mark	Notes
(a)	B1	For using correct Pythagoras theorem to find length *AC or ½ AC*, explicit substitution must be seen.
	M1	For correct trigonometry using their AC in triangle VAC or half of their AC in triangle VAM or correct Pythagoras theorem (let M be the centre of the base at which the diagonals *AC* and *BD* meet) to find length *VA*. *e.g.* $VA = \sqrt{5}x\cos(45°) = \frac{\sqrt{10}}{2}x$ *or* $VA = \frac{\sqrt{5}}{2}x \div \cos(45°) = \frac{\sqrt{10}}{2}x$ *or* $VA = \frac{\sqrt{5}}{2}x \div \sin(45°) = \frac{\sqrt{10}}{2}x$
	A1 cso	Achieves printed answer with no errors. Condone AV instead of VA Note: $\frac{\sqrt{10}}{2}x$ *written as* $\frac{\sqrt{10x}}{2}$ *is A0*, $\frac{\sqrt{10}}{2}x$ *written as* $\frac{\sqrt{10}x}{2}$ *is A1*
(b)	M1	Uses the correct Pythagoras theorem or correct trigonometry with the given length VA= $\frac{\sqrt{10}}{2}x$ to find the height of the pyramid.
	A1	For $h = \frac{\sqrt{5}}{2}x$ Correct exact answer only scores M1A1. This can be written down as the triangle is isosceles.
(c)	M1	**Ignore missing *x* consistently throughout their solution.** For using any appropriate trigonometry to find the angle *VBA* If attempts cosine rule, look for correct application of the cosine rule for the required angle: $\left(\frac{\sqrt{10}}{2}x\right)^2 = \left(\frac{\sqrt{10}}{2}x\right)^2 + (2x)^2 - 2\times\frac{\sqrt{10}}{2}x\times 2x\cos(VBA)$ oe
	A1	For the correct angle. awrt 50.8 ° (with or without degree sign)
(d)	B1	Identifies the angle required.(can be implied/embedded in their working)
	B1	For the correct lengths AM and MB **or** DM and MC **or** BE and ME Let E to be mid point of AB, correct length BE=*x*, ME=0.5*x* scores B1, this mark can be implied/embedded in their working)
	M1	**Ignore missing *x* consistently throughout their solution.** For using any appropriate trigonometry to find the required angle or half of the required angle.
	A1	For the correct angle. awrt 126.9° (with or without degree sign)
(e)	M1	For using the correct formula for the volume of a pyramid with their height of the pyramid and attempts to find a value of *x*.
	A1	For $x = 3$ with or without unit, ignore incorrect unit

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Question	Scheme	Marks

Type MS · Angles and areas of triangles inside a solid

Answer 5 of 8 · Question 4 part b · 3 marks

2020 JAN · Paper 02 (regional R) · extract 1 of 3 · find part b in this mark scheme extract

4 (a)	$A = \frac{1}{2} \times 8 \times 8 \times \sin 60^\circ = \left(16\sqrt{3}\right)$ $48\sqrt{3} = \frac{1}{3} \times '16\sqrt{3}' \times h \Rightarrow h = 9$ *	M1 M1 A1cso [3]
(b)	$BX = \sqrt{9^2 + 8^2} = \sqrt{145}$ $\angle BXC = \frac{'145' + '145' - 8^2}{2 \times '\sqrt{145}' \times '\sqrt{145}'} = 38.8025...^\circ \approx 38.8^\circ$	M1 M1A1 [3]
(c)	Let midpoint of BC be M $AM = \sqrt{8^2 - 4^2} = \left(4\sqrt{3}\right)$ or $MX = \sqrt{145 - 4^2} = \left(\sqrt{129}\right)$ $\angle XMA = \tan^{-1}\left(\frac{9}{'4\sqrt{3}'}\right)$ or $\angle XMA = \sin^{-1}\left(\frac{9}{'\sqrt{129}'}\right)$ or $\angle XMA = \cos^{-1}\left(\frac{'4\sqrt{3}'}{'\sqrt{129}'}\right)$ $= 52.4109...^\circ \approx 52.4^\circ$	M1 M1 A1 [3]
	ALT 1	
	Let midpoint of BC be M $(XA)^2 = (AM)^2 + (XM)^2 - 2(AM)(XM)\cos\theta$ $9^2 = \left(8^2 - 4^2\right) + \left(8^2 + 9^2 - 4^2\right) - 2\sqrt{8^2 - 4^2}\sqrt{8^2 + 9^2 - 4^2}\cos\theta$ $\theta = 52.4109...^\circ \approx 52.4^\circ$	{M1} {M1} {A1} [3]
		Total 9 marks

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ALT	
M1	Use of cosine rule using AX, AM and XM e.g. $(XA)^2 = (AM)^2 + (XM)^2 - 2(AM)(XM)\cos\theta$
M1	Correct values substituted into the cosine rule in any form e.g. $9^2 = (8^2 - 4^2) + (8^2 + 9^2 - 4^2) - 2\sqrt{8^2 - 4^2}\sqrt{8^2 + 9^2 - 4^2}\cos\theta$
A1	awrt 52.4°

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Question number	Scheme	Marks

Type MS · Angles and areas of triangles inside a solid

Answer 6 of 8 · Question 3 part b · 3 marks

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3(a)		
	$AM = \sqrt{10^2 - 8^2}$, $= 6$	M1,A1 (2)
(b)		
	$\cos C = \frac{26^2+16^2-26^2}{2\times16\times26} = \frac{256}{832}$ $\left(=\frac{4}{13}\right.$ oe$\left.\right)$	M1A1
	$\angle BCD = 72°$	A1cao (3)
(c)	$AD = \sqrt{26^2 - 10^2} = 24$ or $DM = \sqrt{26^2 - 8^2} = 6\sqrt{17}$ oe (24.73...)	M1A1
	$\tan(\angle DMA) = \frac{24}{6}$ or $\cos(\angle DMA) = \frac{6}{6\sqrt{17}}$ or $\sin(\angle DMA) = \frac{24}{6\sqrt{17}}$	M1
	$\angle DMA = 76°$	A1cao (4) **[9]**

(a)	
M1	Use Pythagoras, with a minus sign, to obtain the length of AM.
A1	Correct length obtained.
NB	Answers w/o working get both marks (use of (3,4,5) triangle)
(b)	
M1	Use the cosine rule in ΔBCD to obtain a numerical expression for cos C. Correct formula in either form may be quoted and substitution attempted or correct formula can be implied by the correct substitution. Complete method needed, so if another angle found first do not award this mark until a value for angle BCD is obtained.
A1	Correct value for the cosine obtained (Decimal to be awrt 0.308 (0.30769...)) Award by implication if final answer is awrt 72°.
A1cao	For 72° (from correct working)
ALT	**Use the isosceles triangle**
M1A1	$\cos C = \frac{8}{26}$ oe (Any trig function allowed provided work completed to a value for angle BCD)
A1cao	For 72° (from correct working)
(c)	
M1	Attempt the length of AD or DM using Pythagoras with a minus sign.
A1	Correct value for their choice of line,
M1	Use an appropriate trig function. The length of AD or DM must have been attempted with a + or a - sign.
A1cao	Correct answer.
	Penalise once only in (b) and (c) for failing to round as instructed.

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Answer 6 of 8 · Question 3 part b · 3 marks

2020 JAN · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

Question Number	Scheme	Marks
	$\longrightarrow$ $\longrightarrow$	

Type MS · Angles and areas of triangles inside a solid

Answer 7 of 8 · Question 8 part c · 3 marks

2025 JUN · Paper 01 · extract 1 of 4 · find part c in this mark scheme extract

8(a)	$\tan 30° = \frac{VP}{6}$ where P is the centre of the base $VP = 6\tan 30° = 2\sqrt{3}$ *	M1 A1*cso [2]
(b)	$AC = \sqrt{12^2 + 12^2} = 12\sqrt{2}$ or $AC = \frac{12}{\sin 45}$ or $AC = \frac{12}{\cos 45}$ $VA^2 = \left(2\sqrt{3}\right)^2 + \left(6\sqrt{2}\right)^2 (= 84)$ $VA = 2\sqrt{21}$ (cm) oee	M1 oe M1 oe A1 [3]
(c)	$VX = \sqrt{\left('2\sqrt{21}'\right)^2 - 6^2} \left(= 4\sqrt{3}\right)$ or $VX = \frac{6}{\cos 30}$ or $VX = \sqrt{6^2 + \left(2\sqrt{3}\right)^2}$ where X is the midpoint of the side AD Area $= \frac{1}{2} \times 4\sqrt{3} \times 12 = 24\sqrt{3}$ (cm^2) oee	M1 M1A1 [3]
(d)	$'24\sqrt{3}' = \frac{1}{2} \times '2\sqrt{21}' \times DY$ where Y is the foot of the perpendicular from D to VA $DY = \frac{24\sqrt{7}}{7}$ oee	M1 A1 [2]
(e)	$\cos\theta = \frac{\left(,\frac{24\sqrt{7}}{7},\right)^2 + \left(,\frac{24\sqrt{7}}{7},\right)^2 - \left('12\sqrt{2}'\right)^2}{2 \times ,\frac{24\sqrt{7}}{7}, \times ,\frac{24\sqrt{7}}{7},} = -\frac{3}{4}$ $\theta = 138.6°$	M1A1 A1 [3]
		Total is 13 marks

Type MS · Angles and areas of triangles inside a solid

Answer 7 of 8 · Question 8 part c · 3 marks

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Part	Mark	Additional Guidance
(a)	M1	For forming a correct equation using trigonometry. e.g. $\tan 60° = \frac{6}{VP}$ or $\frac{h}{\sin 30°} = \frac{6}{\sin 60°}$ (If attempted by ratio, send to review)
	A1*cso	For **rearranging** to show the required result. Minimum need to see one intermediate step from their equation to the answer ,if the height is not the subject of their equation: e.g. $\tan 30° = \frac{VP}{6} \Rightarrow VP = 6\tan 30° = 2\sqrt{3}$ is M1A1 $VP = 6\tan 30° = 2\sqrt{3}$ is M1A1 $\tan 30° = \frac{VP}{6} \Rightarrow \frac{\sqrt{3}}{3} = \frac{VP}{6} \Rightarrow VP = 2\sqrt{3}$ is M1A1 $\tan 30° = \frac{VP}{6} \Rightarrow VP = 2\sqrt{3}$ is M1A0
(b)	M1oe	Correct use of Pythagoras' theorem or trigonometry to obtain the length/half length of the diagonal of the base AC or $\frac{1}{2}AC$ Or other valid method to find VM, (Let M to be the midpoint of AB) e.g.: $VM = \frac{6}{\cos 30}\left(=4\sqrt{3}\right)$ or $VM = \sqrt{6^2 + \left(2\sqrt{3}\right)^2}\ \left(=4\sqrt{3}\right)$
	M1oe	Use Pythagoras' theorem to obtain a numerical value for VA / VA^2 Any correct method. e.g. $VA^2 = \left(4\sqrt{3}\right)^2 + (6)^2$
	A1	Correct exact length $2\sqrt{21}$ or $\sqrt{84}$ oee (or exact equivalent)
(c)	M1	Correct use of Pythagoras' theorem or trigonometry to obtain the length of VX, which is the same as VM, could be found in part b Or correct use of cosine rule (ft their VA) to find the value of cosine value of any angle in triangle VAD e.g. $AD^2 = VA^2 + VA^2 - 2VA \times VA \times \cos\angle AVD$ $12^2 = \left('2\sqrt{21}'\right)^2 + \left('2\sqrt{21}'\right)^2 - 2\times'2\sqrt{21}'\times'2\sqrt{21}'\times\cos\angle AVD$ $\left\{\Rightarrow \cos\angle AVD = \frac{1}{7}\right\}$
	M1	Correct method to find the area of triangle VAD (Do not have to use exact values) Accept any correct valid method. e.g. $\left\{\cos\angle AVD = \frac{1}{7} \Rightarrow \sin\angle AVD = \frac{4\sqrt{3}}{7}\right\} \Rightarrow A = \frac{1}{2}\times'2\sqrt{21}'\times'2\sqrt{21}'\times'\frac{4\sqrt{3}}{7}' = 24\sqrt{3}$
	A1	Correct **exact** value for area
(d)	M1	Use of area of triangle with VA as the base to form an equation for DY Or other valid method: e.g. using trigonometry ratio: $\sin\angle VAD = \frac{'4\sqrt{3}'}{'2\sqrt{21}'} = \frac{DY}{12}$

Type MS · Angles and areas of triangles inside a solid

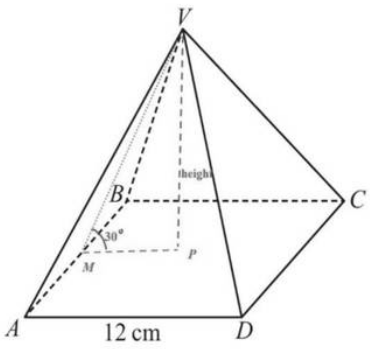
Answer 7 of 8 · Question 8 part c · 3 marks

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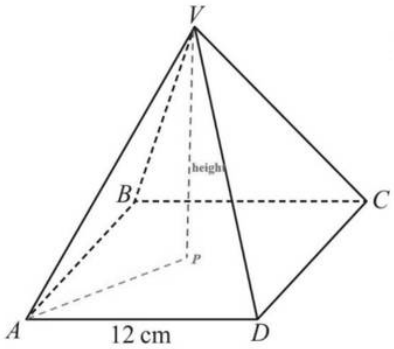
	A1	Correct exact length $DY = \frac{24\sqrt{7}}{7}$ oee, e.g. $DY = \frac{8\sqrt{63}}{7}$ or $24\sqrt{\frac{1}{7}}$
(e)	M1	Correct use of cosine rule: $\cos\theta = \frac{"DY"^2 + BY"^2 - "BD"^2}{2\times"DY"\times"BY"}$ with their values Or other valid method, e.g. $\sin\frac{\theta}{2} = \frac{"\frac{BD}{2}"}{"DY"} = \frac{"6\sqrt{2}"}{"\frac{24\sqrt{7}}{7}"}$
	A1	Obtains $\cos\theta = -\frac{3}{4}$ or $\sin\frac{\theta}{2} = \frac{\sqrt{14}}{4}$
	A1	Correct angle found

Useful diagrams:

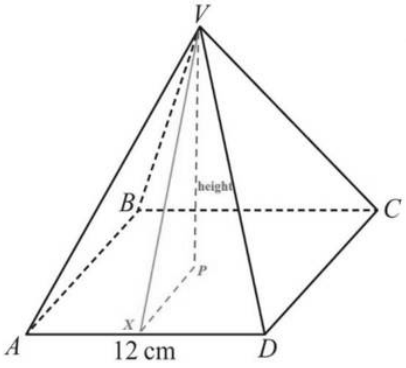
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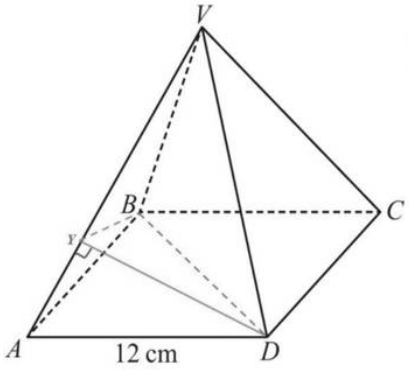
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3



4



Type MS · Angles and areas of triangles inside a solid

Answer 7 of 8 · Question 8 part c · 3 marks

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Question number	Scheme	Marks

Type MS · Angles and areas of triangles inside a solid

Answer 8 of 8 · Question 9 part c · 5 marks

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9(a)	$AC = \sqrt{12^2+12^2} = \sqrt{288} = 12\sqrt{2}$	M1A1 [2]
(b)	$x = \dfrac{\dfrac{12\sqrt{2}}{2}}{\cos 30^\circ} = \sqrt{96} = \left(4\sqrt{6}\right)*$	M1A1 cso [2]
(c)	$\cos \angle AOB = \dfrac{96+96-12^2}{2\times\sqrt{96}\times\sqrt{96}} \Rightarrow \angle AOB = 75.522...^\circ$ Area $\Delta OAB = \dfrac{1}{2}\times\sqrt{96}\times\sqrt{96}\times\sin'75.522^\circ{}' = (46.4758...)$ $4\times 46.4758+12^2 = 329.90.... \Rightarrow$ Area = 330 [m^2]	M1A1 M1 M1A1 [5]
(d)	$\left[\angle OBC = \dfrac{180-75.522}{2} = 52.239^\circ\right]$ Let Y on OB be the foot of the perpendicular from A to OB $AY = 12\sin 52.239^\circ = (9.4868...)$ $\cos \angle AYC = \dfrac{'9.4868'^2+'9.4868'^2-288}{2\times'9.4868'\times'9.4868'} \Rightarrow \angle AYC = 126.87... \approx 127^{[o]}$	M1 A1 M1A1 [4]
		Total 13 marks

Question	Notes	Marks
9(a)	For using Pythagoras theorem on triangle ABC or triangle ADC $AC = \sqrt{12^2+12^2} = ...$	M1
	For the correct value of AC $AC = \sqrt{288} = 12\sqrt{2}$	A1 [2]
(b)	Let the intersection of AC and BD be X Uses any appropriate trigonometry on triangle OAX For example; $x = \dfrac{\dfrac{12\sqrt{2}}{2}}{\cos 30^\circ} = ...$ **OR** $\dfrac{\sin 120^\circ}{12\sqrt{2}} = \dfrac{\sin 30^\circ}{OA} \Rightarrow OA = ...$ OR $\left(12\sqrt{2}\right)^2 = x^2+x^2-2\times x\times x\cos 120 \Rightarrow x = ...$	M1
	For the correct value of x	A1 cso

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Answer 8 of 8 · Question 9 part c · 5 marks

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	$x = \sqrt{96} = \left(4\sqrt{6}\right)$ *	[2]
(c)	For using trigonometry to find angle $\angle AOB$ $\cos \angle AOB = \frac{96+96-12^2}{2\times\sqrt{96}\times\sqrt{96}} = ...$	M1
	$\angle AOB = 75.522...^{\circ}$	A1
	Area of triangle OAB Area $= \frac{1}{2}\times\sqrt{96}\times\sqrt{96}\times\sin'75.522^{\circ}' = \left(46.4758...\right)$	M1
	Total area of the pyramid $= 4\times 46.4758 + 12^2 = \left(329.90....\right)$	M1
	For the correct final area = awrt 330 (cm^2)	A1 [5]
	ALT	
	Height of the triangle of one of the triangular faces: $h = \sqrt{\left(4\sqrt{6}\right)^2 - 6^2} = ...$	M1
	$h = 2\sqrt{15}$	A1
	Area of triangle $OAB = \frac{1}{2}\times 2\sqrt{15}\times 12 = 12\sqrt{15} = \left(46.4758...\right)$	M1
	Total area of the pyramid $= 4\times 12\sqrt{15} + 144 = \left(329.903...\right)$	M1
	For the correct final area = awrt 330 (cm^2) Allow an exact answer of $48\sqrt{15}+144$ $(cm)^2$ oe	A1
(d)	$\cos \angle AOB = \frac{96+96-12^2}{2\times\sqrt{96}\times\sqrt{96}} = 75.522^{\circ}$ or $\angle OBC = \frac{180^{\circ} - 75.522^{\circ}}{2} = 52.239^{\circ}$	
	Let Y on OB be the foot of the perpendicular from A to OB Length $AY = 12\sin 52.239^{\circ} = \left(9.4868...\right)$ OR Length $AY = 4\sqrt{6}\sin 75.522^{\circ} = \left(9.4868...\right)$	M1A1 [M1A1]
	For the appropriate trigonometry on triangle AYC to find angle AYC $\cos \angle AYC = \frac{'9.4868'^2 + '9.4868'^2 - 288}{2\times'9.4868'\times'9.4868'} \Rightarrow \left(\angle AYC = 126.87\right)$	M1
	Angle between plane AOB and plane OBC =awrt 127[°]	A1 [4]
		Total 13 marks

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Answer 8 of 8 · Question 9 part c · 5 marks

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Question	Scheme	Marks